

Epistemic Immutability

Provenance,
Counterexamples,
and Selective
Continuation

Why an artifact's
metadata records its
history, but does not
establish it.

The diagram features a large rectangular frame. Inside, a horizontal sequence of characters 'a1b2c3d4e5f6...' is displayed. A blue double-headed vertical arrow is positioned to the left of the text, spanning its height. From the right side of the text, two dashed orange lines branch out diagonally upwards and downwards. Each dashed line is followed by a solid orange line that continues horizontally to the right edge of the frame. Small orange 'x' marks are placed at the points where the dashed lines meet the solid lines, and at the junction where the dashed lines branch from the text.

a1b2c3d4e5f6...



Modification Date:
Year 2036

The Fallacy of the Untampered Record

- A timestamp is not a report of an event; it is a property of a record.
- A syntactically valid but impossible timestamp preserved perfectly by a computer is genuine evidence—just not of the event it appears to describe.
- The reflex is to overwrite this ‘garbage’ data. This is an epistemic error.

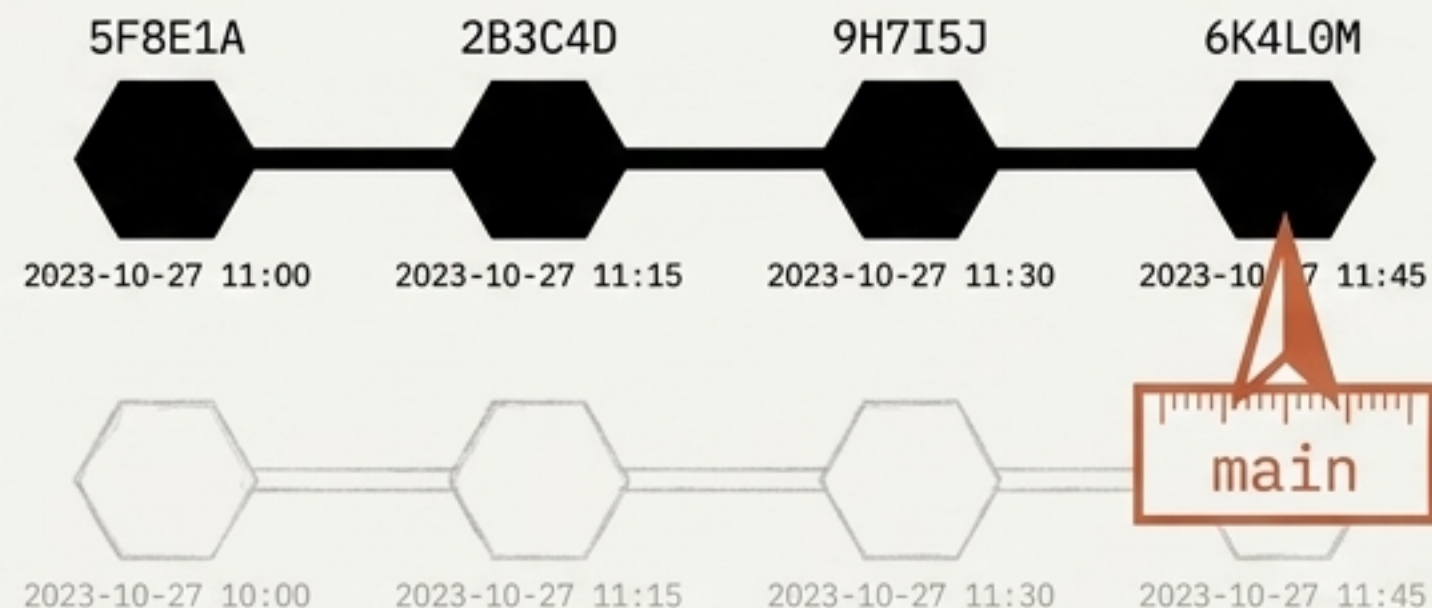
Unchanged $\neq$ True

The Git DAG Illusion



State 1: Original DAG

main



State 2: Rewritten History

main

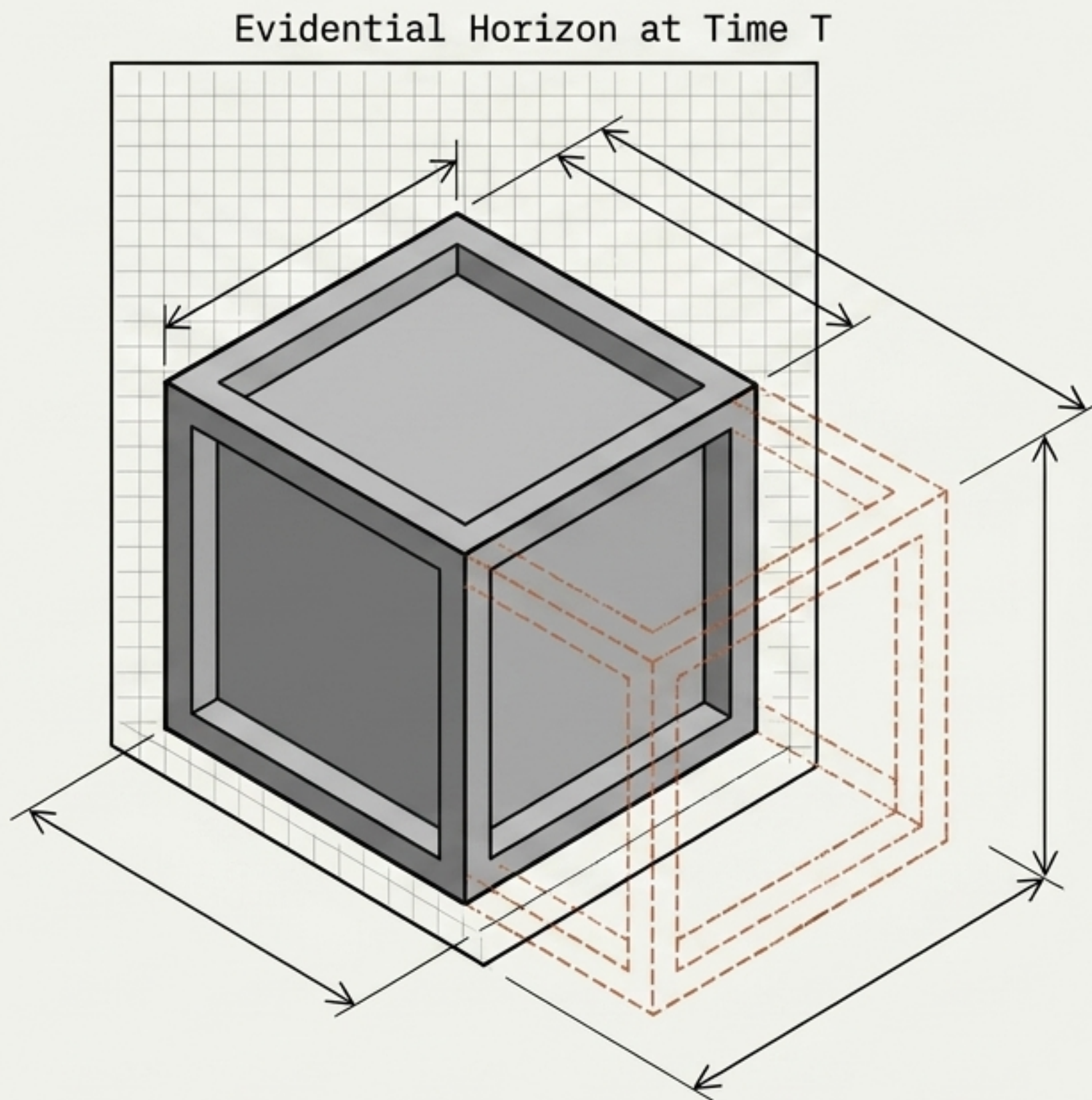
Object Immutability

Pointer Reassignment

Git: Immutable Objects, Revisable Ancestry

Amending a commit date and force-pushing constructs a parallel history and moves a pointer. It does not travel backward to rewrite the past. Objects are genuinely immutable; ancestry is fully revisable.

Integrity preserves what was asserted. It does not establish that the assertion was true when made.



ML Predictions as Bounded Commitments

- A vision model predicts the unseen side of an object. Later, the object rotates, revealing the true back. The prediction is falsified.
- The Flawed Reflex: Discard the old prediction and save only the new, correct result.
- The Reality: The mistaken prediction is the experiment. Without a retained record of what the model committed to before additional evidence arrived, we cannot test how it integrates new information.

Do not overwrite the falsified output.

Three Scales, One Universal Structural Error

Domain	The Artifact	The Claim	The Flawed Reflex	The Lost Evidence
Audio	Sony Recording	"Recorded in 2036"	Silently fix the timestamp	Device clock drift hidden
Version Control	Authored Commit	"Authored 3 years ago"	Force-push the rewritten DAG	True timeline of work erased
Machine Learning	Bounding Box	"The back looks like X"	Overwrite with ground truth	Model miscalibration un-testable

At every scale, the reflex to 'overwrite with the truth' destroys the evidence that made the correction possible.

The Three Geometries of History

Layer 1: Declared History

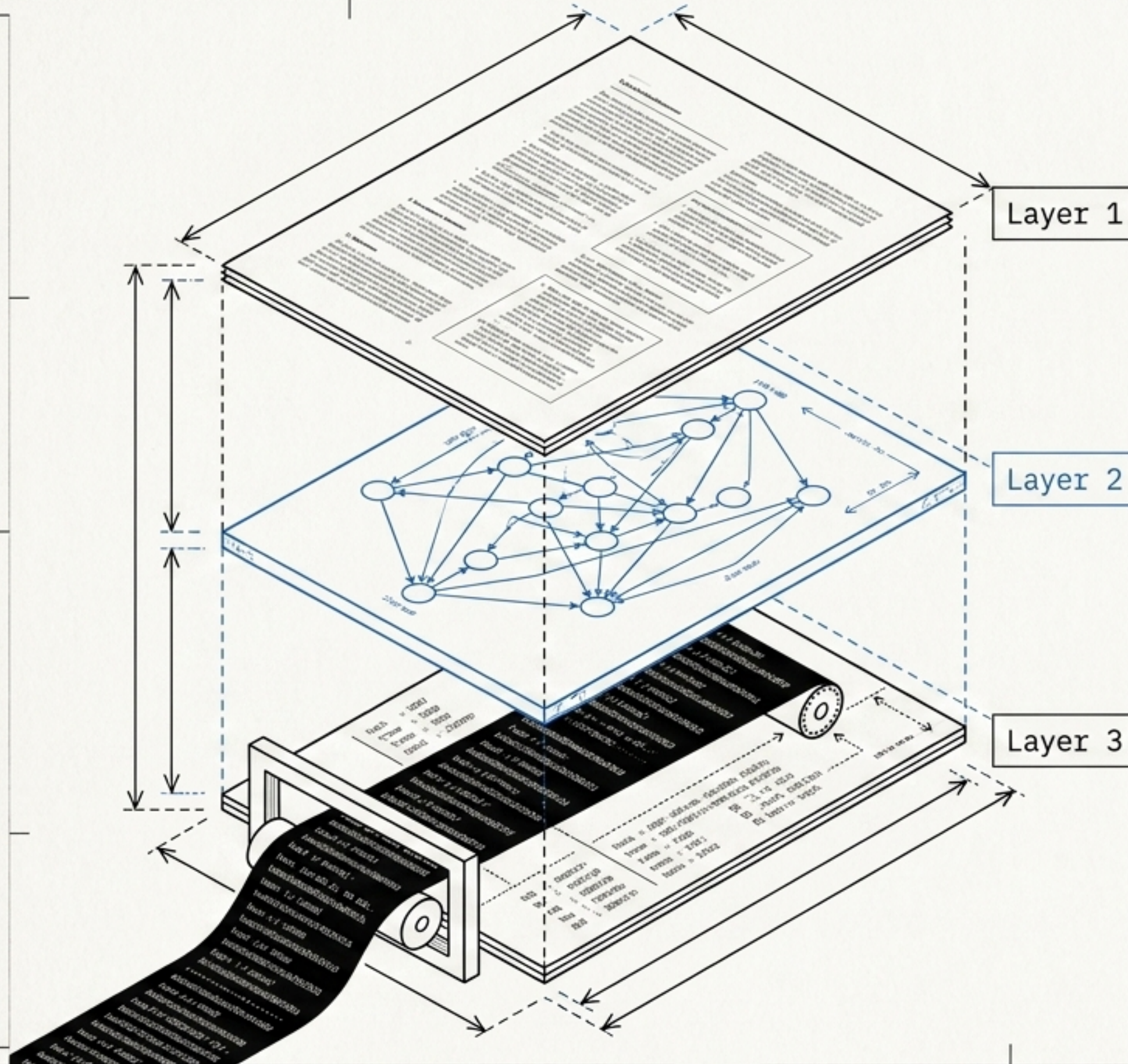
The artifact's self-account (e.g., self-reported timestamps). Governed purely by syntax.

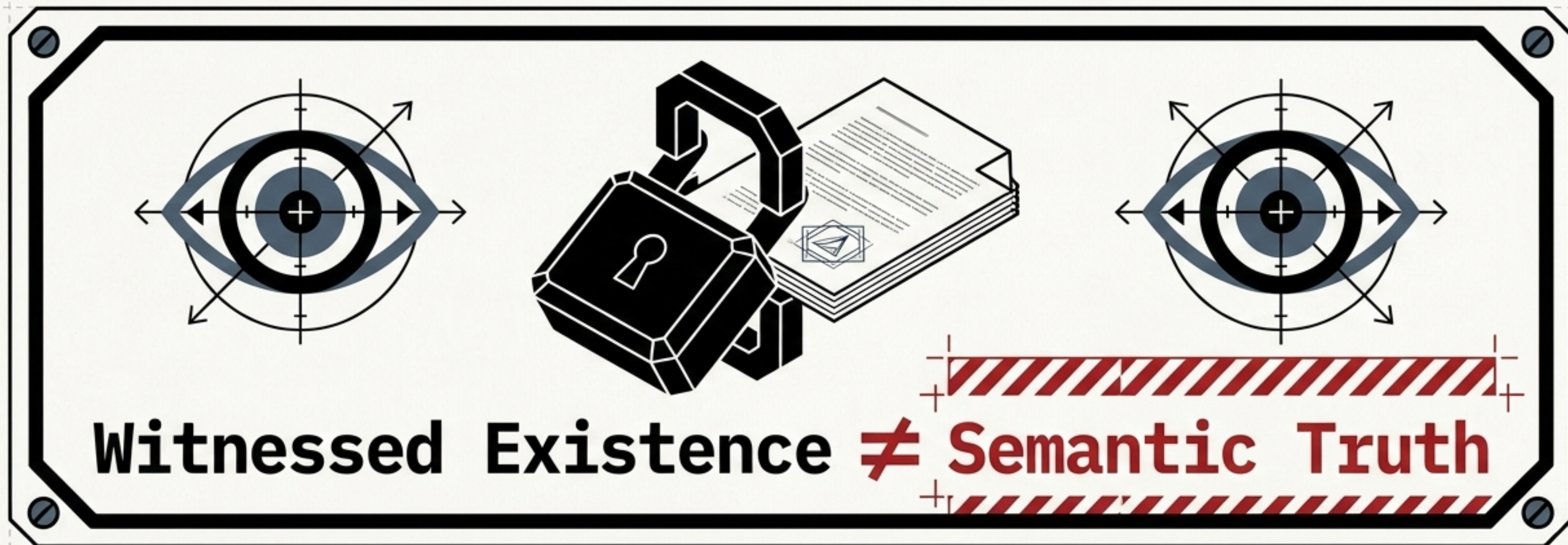
Layer 2: Structural History

The graph of causal precedents (e.g., Git parent pointers). Governed by cryptography.

Layer 3: Witnessed History

Independent proof of existence. Governed by external transparency logs.





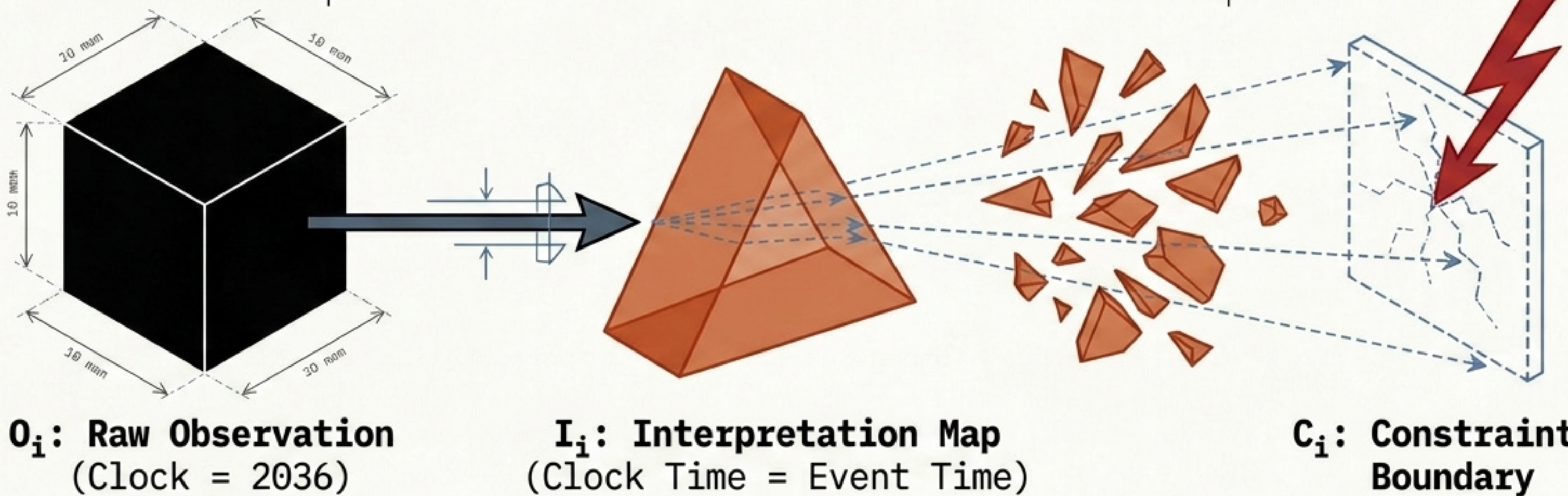
The Strict Limits of the Witness Layer

Append-only external logs prove exactly one thing: an artifact existed, unaltered, at a given time.

They do not certify any assertion inside the document. The witness layer supplies timing, not truth. A well-witnessed record is not automatically a verified one.

Formalizing Evidence: Observation vs. Interpretation

$$\text{Constraint } C_i = \phi(O_i, I_i, \tau_i)$$



Contradictory evidence implicates the hidden assumption (the interpretation map), not the raw observation.

If we preserve O_i and I_i separately, we protect the **exact data** from the flawed assumption.

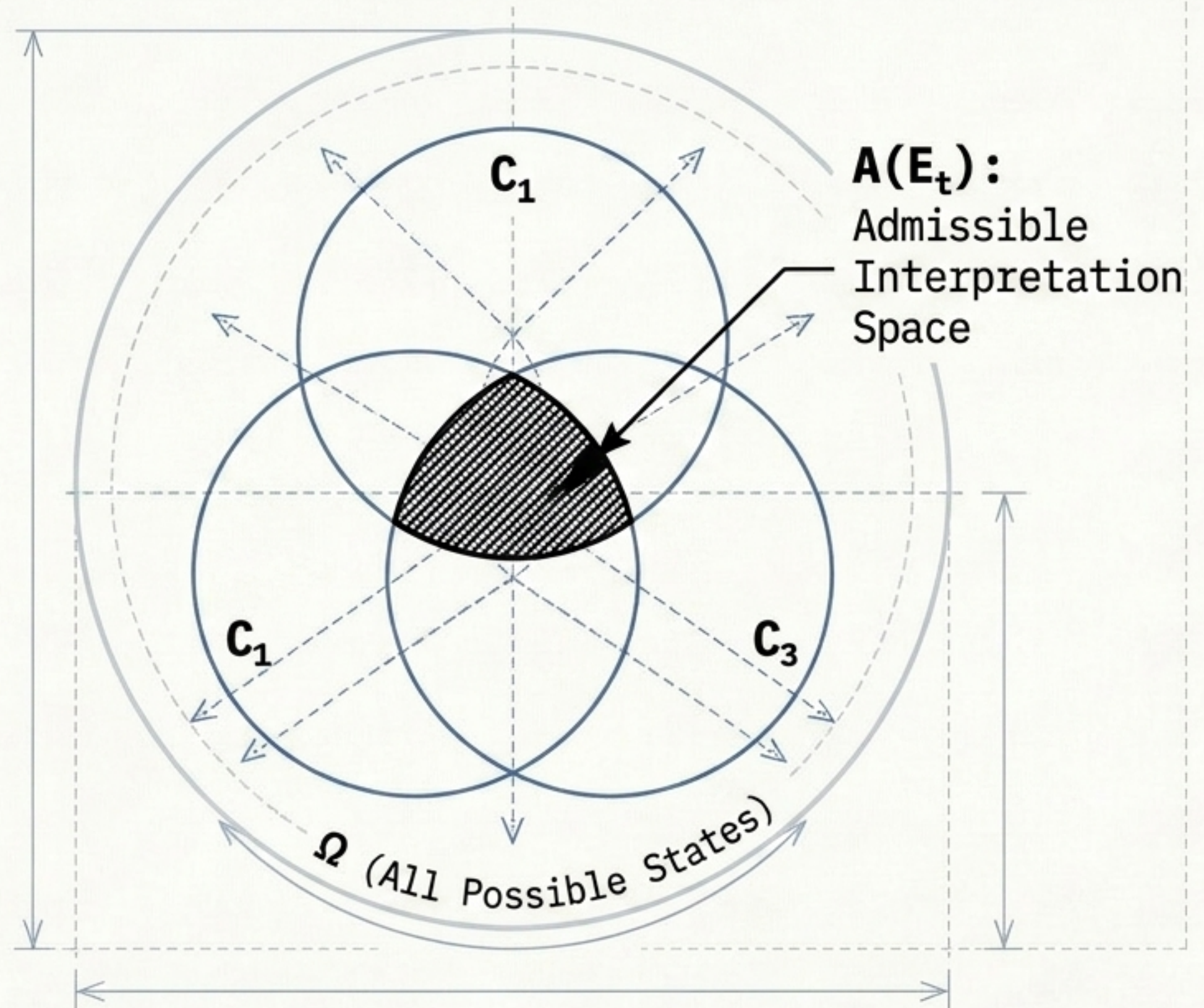
Constraining the Admissible Space

$$A(E_t) = \bigcap_{i \leq t} C_i$$

As evidence accumulates, the space narrows via intersection.

Monotonicity

When evidence is genuinely appended rather than revised, the space only shrinks. It never forgets what was previously excluded.



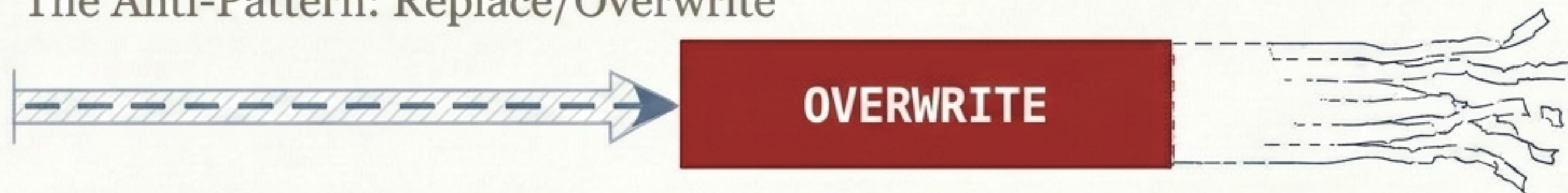
Expansion, Contradiction, and False Exclusion

Failure Mode	Mathematical Definition	Symptom	Severity
Expansion	$A(E_{t+1}) \not\subseteq A(E_t)$	New evidence fails to constrain old space.	Low
Contradiction	$A(E_{t+1}) = \emptyset$	Empty intersection / hidden assumption fails.	Medium
False Exclusion	$\omega^* \notin A(E_{t+1})$ while $A(E_{t+1}) \neq \emptyset$	The true state is excluded but the system remains confident and internally consistent.	Critical

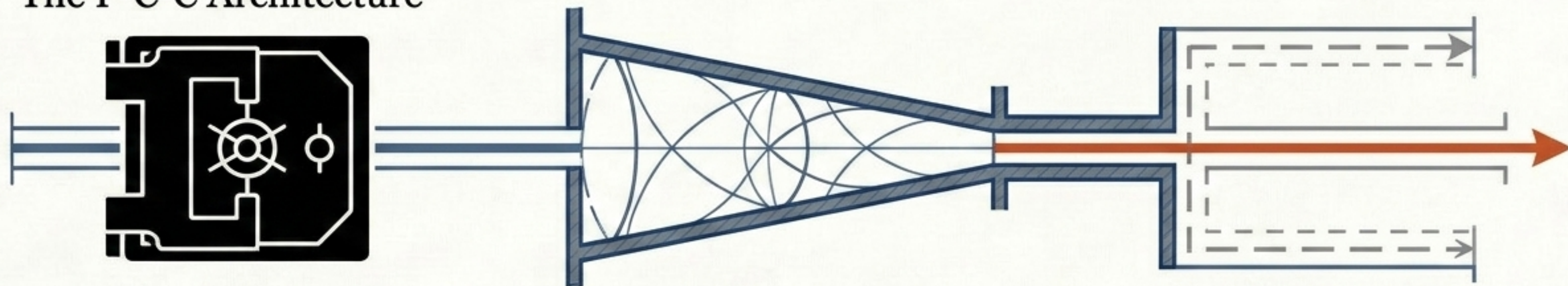
False exclusions are latent. They are only diagnosed **retroactively**—which is why old commitments must be preserved.

The Core Framework: Preserve, Constrain, Continue

† The Anti-Pattern: Replace/Overwrite †



The P-C-C Architecture



Preserve (P_t):
Retain original
commitment unmodified.

Constrain (A_{t+1}):
Update interpretation
space with new evidence.

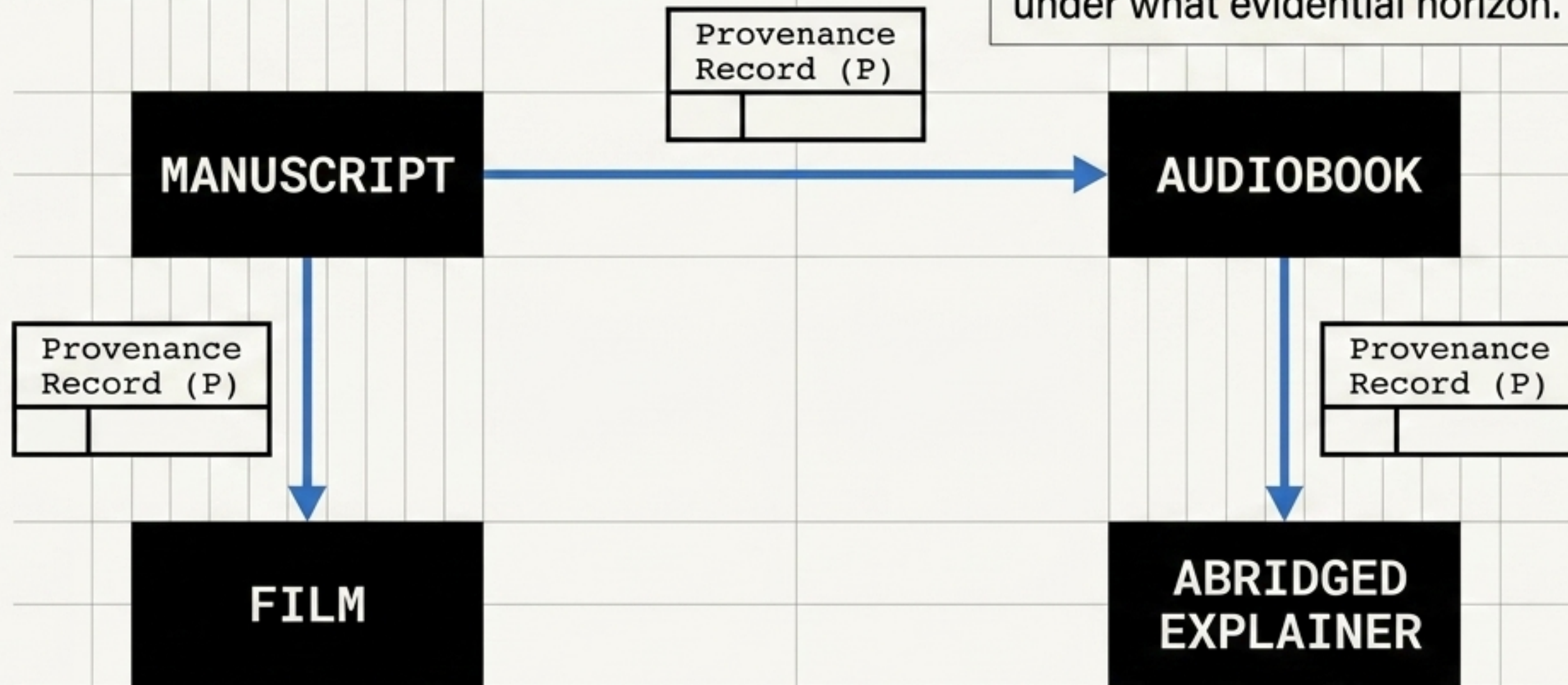
Continue (H_t):
Choose which admissible
branch to build upon.

Append the correction, record the discrepancy, correct the continuation.

Constructive Architecture: Media Quines

Artifacts designed to carry their own provenance across every transformation applied to them.

An interoperable directed graph where each transformation retains exactly how it was produced, from what inputs, under what evidential horizon.



Provenance Graphs & Preservation Contracts



We do not demand pixel-perfect identity across transformations. An audiobook and a simulation should not be identical.

The Contract (J_e):

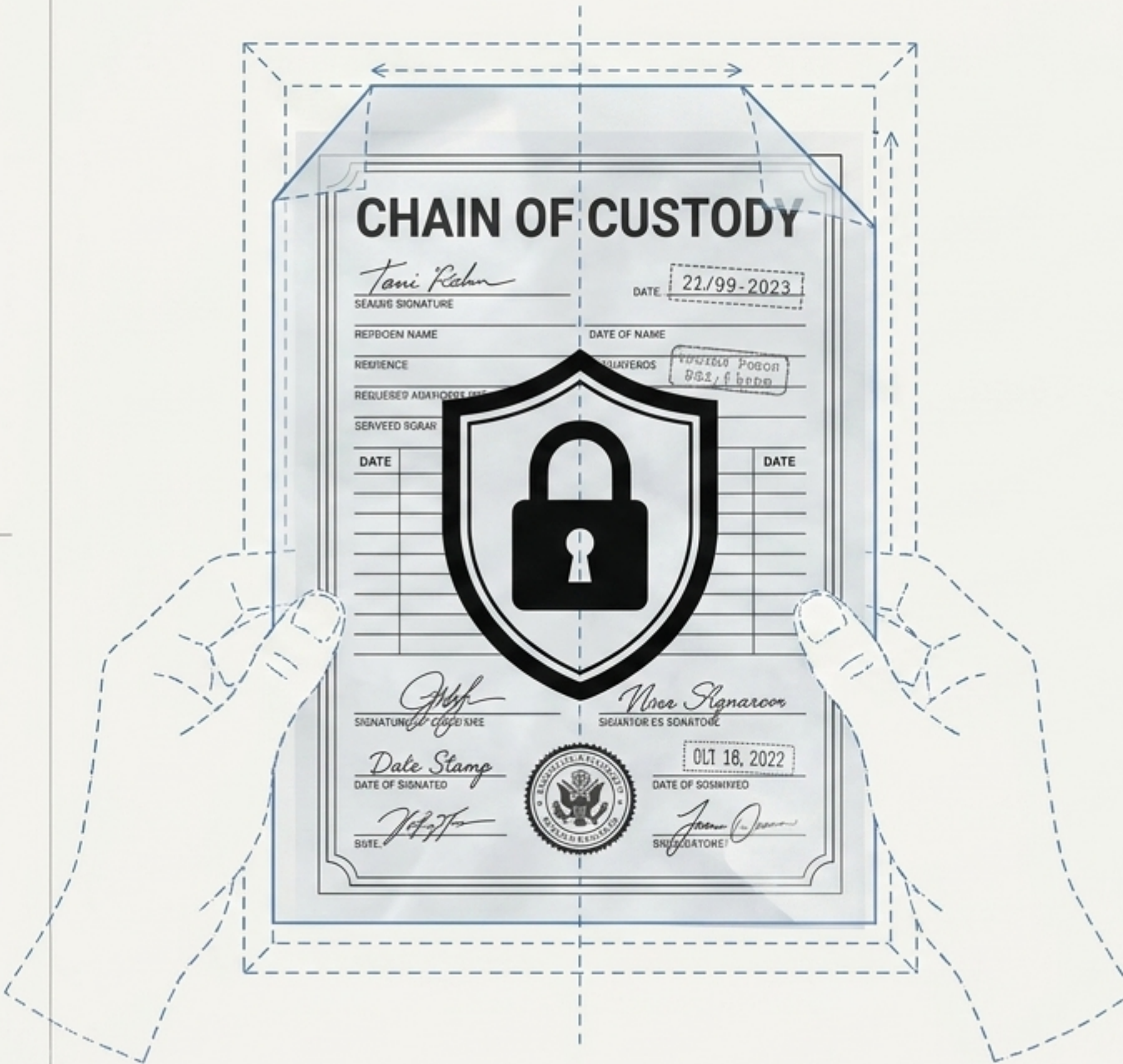
Each transformation must explicitly state which mathematical invariants it claims to carry forward.

The Test:

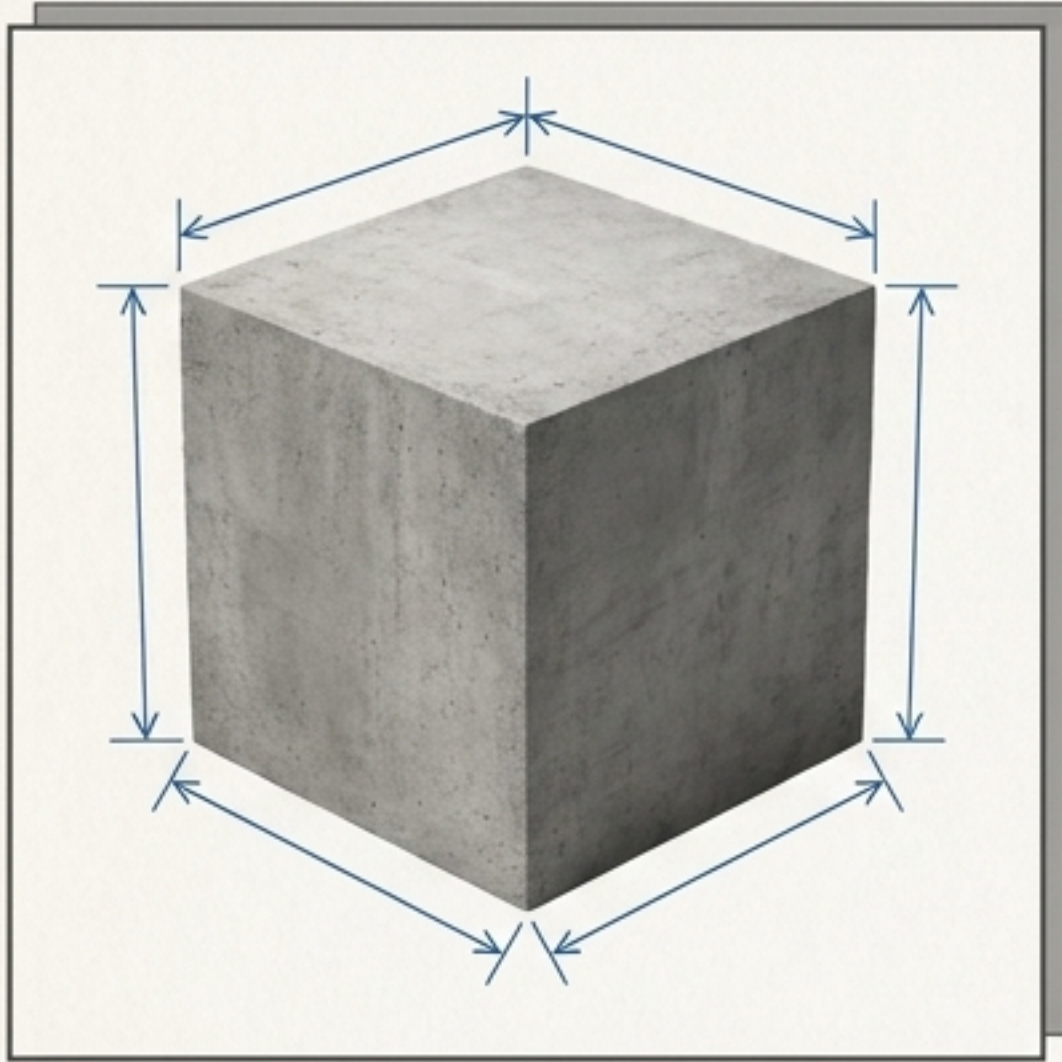
We evaluate disparate outputs by checking if the invariant they both promised to preserve actually agrees.

The Illusion of Declared Provenance

- Internal metadata asserting ownership, severity, or chain-of-custody has the exact same epistemic status as a broken clock.
- Versioning and hashing these fields only proves they haven't been altered; it does not make them true.
- **An evidentiary vocabulary attached to a value nobody has actually witnessed is an epistemic ghost.**

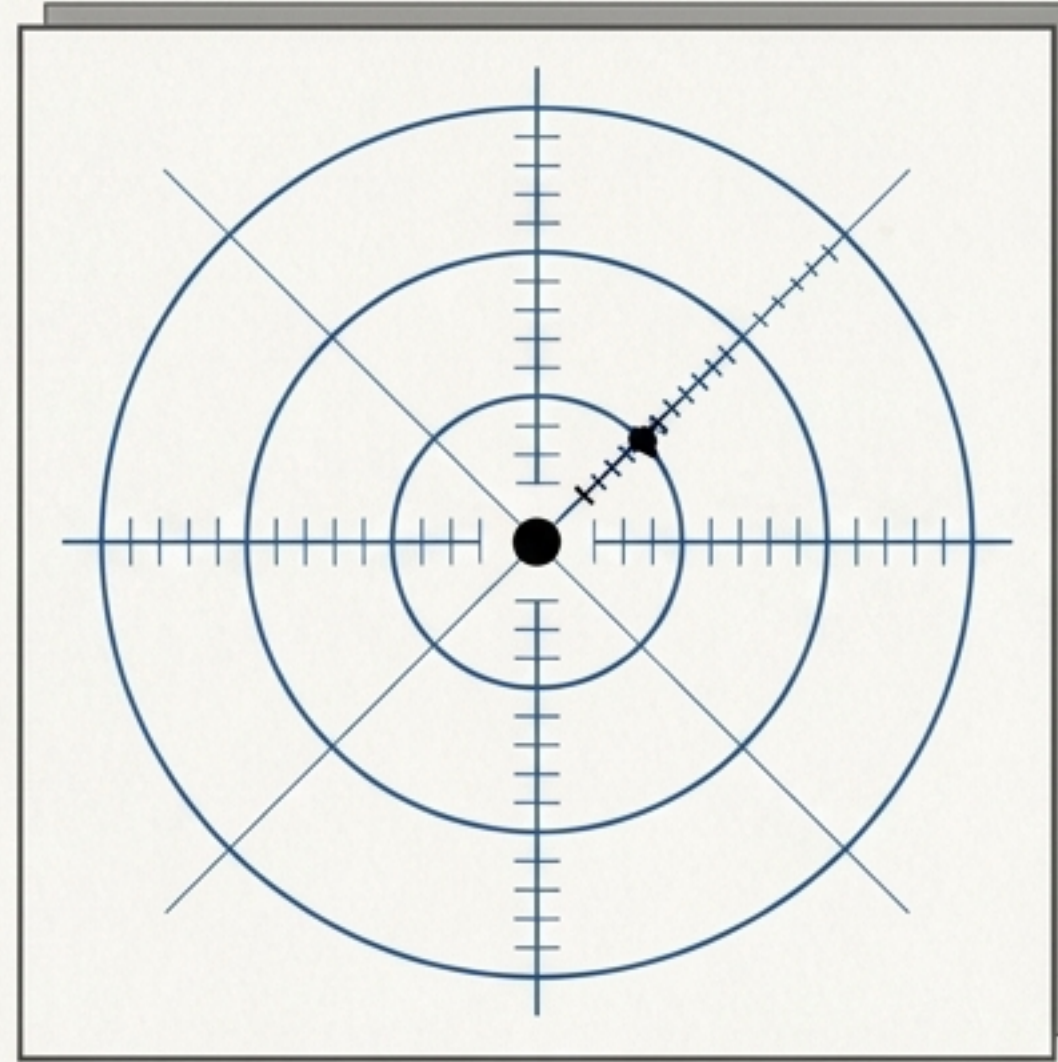


A New Discipline for Epistemic Artifacts



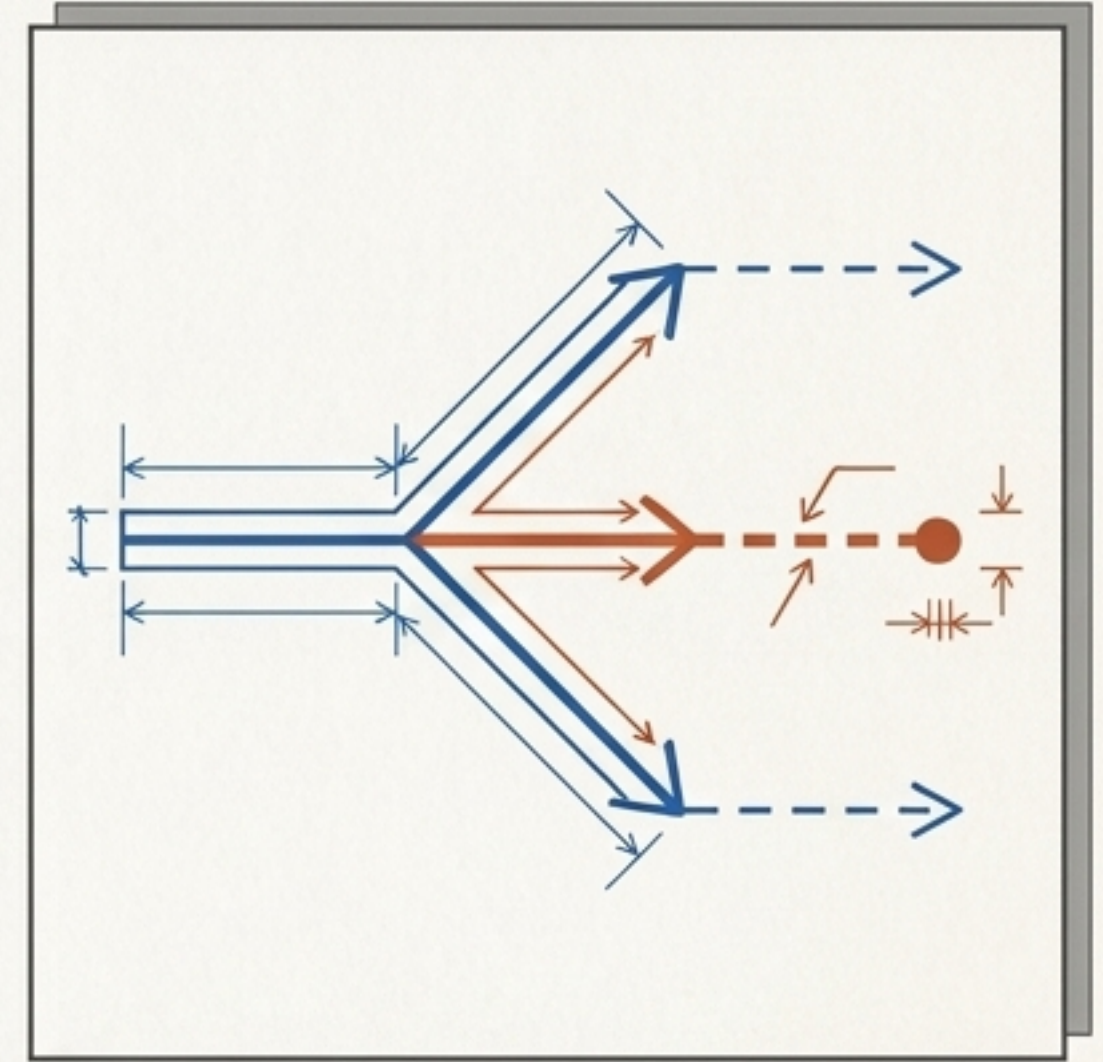
Preservation without correction

History is Monotonic



Constraint without erasure

Belief is Revisable



Continuation without canonicalization

Commitments are Falsifiable

Falsified predictions, broken clocks, and rewritten commits are not mistakes to be cleaned out of the historical record. They are the record functioning exactly as it should.